AP CALCULUS AB · EXAMINATION REPOSITORY

Unit 1
Limits & Continuity

100-Question Master Examination

- Questions 1–90: Multiple Choice
- Questions 91–100: Free Response with Graphs and Tables
- Complete Answer Key and Solutions at the End

Examination Structure

This paper contains 100 continuously numbered questions. All answers, worked solutions, and scoring rubrics appear at the end of the document.

⭐ Priority Rating System

★★★★★	**Very Important**	High-frequency exam concepts: IVT, Squeeze Theorem, continuity definition, asymptotes, one-sided limits.
★★★	**Mandatory**	Essential skills: factoring and conjugates, piecewise parameters, limits at infinity, special trig limits.
★★	**Important**	Foundational practice: direct substitution, basic table reading, simple rational limits.

- **Questions 1–90:** Multiple choice, four options, exactly one correct answer.
- **Questions 91–100:** Free response, 9 points each, with graphs and tables where appropriate.
- **Difficulty Mix:** Easy 5 · Moderate 68 · Hard 17.
- **Calculator Policy:** No calculator required. Question 100 permits rounded decimal values.
- **Answer Section:** Begins after Question 100 — answer key, then full MCQ solutions, then FRQ solutions with rubrics.

Topic Coverage (Unit 1.1 – 1.16)

1.1 Change at an Instant

1.2 Limit Notation

1.3–1.4 Graphical and Tabular Limits

1.5 Algebraic Limit Properties

1.6 Algebraic Manipulation

1.7 Selecting Procedures

1.8 Squeeze Theorem

1.9 Multiple Representations

1.10 Types of Discontinuity

1.11 Continuity at a Point

1.12 Continuity on an Interval

1.13 Removing Discontinuities

1.14 Infinite Limits

1.15 Limits at Infinity

1.16 Intermediate Value Theorem

Directions: Attempt the entire paper before consulting the answer section. For free response questions, show all reasoning; unsupported answers earn no credit.

QUESTION 01

EASY · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 2} (x^2 - 4)/(x - 2) =$

A0
B4
C2
DThe limit does not exist

QUESTION 02

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x \to -1} (2x^2 + 3x - 5) =$

A4
B−10
C0
D−6

QUESTION 03

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 3} (x^2 - 2x - 3)/(x^2 - 9) =$

A0
BThe limit does not exist
C2/3
D1

QUESTION 04

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to -2} (x^3 + 8)/(x + 2) =$

A4
B12
C8
D0

QUESTION 05

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 4} (x - 4)/(\sqrt{x} - 2) =$

A0
B4
C8
D1

QUESTION 06

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 0} (\sqrt{(9 + x)} - 3)/x =$

A1/6
B0
C1/3
D1/2

QUESTION 07

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to 1} [1/(x+1) - 1/2]/(x-1) =$

A −1/2
B 1/4
C 1/2
D −1/4

QUESTION 08

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to 0} (x^2 + 4x)/(x^3 - 2x) =$

A 0
B −2
C 2
D The limit does not exist

QUESTION 09

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} (\sqrt{(1+2x)} - (1+x))/x^2 =$

A 0
B 1/2
C The limit does not exist
D −1/2

QUESTION 10

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\lim_{x \to 2} (x^3 - 2x^2 - x + 2)/(x^2 - 4) =$

A0
B3/4
C1
D3

QUESTION 11

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

$\lim_{x \to -3} (2x^2 + 5x - 3)/(x^2 - 9) =$

AThe limit does not exist
B−7/6
C7/6
D1/6

QUESTION 12

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Let $f(x) = \{ x^2 - 1 \text{ for } x < 2; 3x - 2 \text{ for } x \geq 2 \}$. Which statement about $\lim_{x \to 2} f(x)$ is true?

AThe limit exists and equals 3
BThe limit exists and equals 2
CThe limit exists and equals 4
DThe limit does not exist

QUESTION 13

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Let $f(x) = \{ 2x + k \text{ for } x \le 1; x^2 + 3 \text{ for } x > 1 \}$. For what value of k is f continuous at $x = 1$?

A0
B2
C4
D1

QUESTION 14

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x \to 0} \sin^2(3x) / (x \cdot \sin(6x)) =$

A1/2
B0
C3/2
D3

QUESTION 15

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x \to 0} (1 - \cos(3x))/x^2 =$

A0
B3/2
C9/2
D9

QUESTION 16

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 0} x / \tan(5x) =$

A1/5
B1
C5
D0

QUESTION 17

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 0} \sin(4x)/\sin(2x) =$

A4
B1/2
C1
D2

QUESTION 18

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x \to 0} (\tan x - \sin x)/x^3 =$

A0
B1
C1/2
D1/3

QUESTION 19

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to\pi} \sin x / (x - \pi) =$

A0
B−1
CThe limit does not exist
D1

QUESTION 20

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to\pi/2} \cos x / (\pi/2 - x) =$

A0
B−1
Cπ/2
D1

QUESTION 21

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to0} (1 - \cos x)/(x \cdot \sin x) =$

AThe limit does not exist
B1/2
C1
D0

QUESTION 22

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x\to 0} \sin(x^2)/x =$

A1
B∞
CThe limit does not exist
D0

QUESTION 23

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to 0} (1 - \cos^2 x)/x^2 =$

A1
B2
CThe limit does not exist
D0

QUESTION 24

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} |x| \cdot \sin(1/x) =$

AThe limit does not exist
B1
C0
D−1

QUESTION 25

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to 0} (\sec x - 1)/x^2 =$

A1
B0
C1/2
DThe limit does not exist

QUESTION 26

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x\to \infty} (4x^2 - 1)/(x^2 + 3x) =$

A4
B0
C1/4
D∞

QUESTION 27

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to \infty} (3x^3 - x)/(2x^3 + 5x^2) =$

A3
B0
C∞
D3/2

QUESTION 28

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x\to\infty} (x^2 + 1)/(2x + 3) =$

A∞
B1/2
C0
D−∞

QUESTION 29

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to-\infty} (x^2 + 1)/(2x + 3) =$

A−1/2
B−∞
C+∞
D0

QUESTION 30

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to\infty} 2x / \sqrt{(x^2 + 1)} =$

A2
B1
C∞
D0

QUESTION 31

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to-\infty} 2x / \sqrt{(x^2+1)} =$

A2
B−2
CThe limit does not exist
D0

QUESTION 32

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to\infty} (\sqrt{(4x^2+x)} - 2x) =$

A1/4
B1/2
C0
D2

QUESTION 33

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to\infty} (\sqrt{(x^2+3x)} - x) =$

A3
B3/2
C∞
D0

QUESTION 34

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The vertical asymptote(s) of $f(x) = (x^2 - 4)/(x^2 - x - 2)$ is/are

Ax = −1 and x = 2
Bx = −1 only
Cx = 2 only
Dx = −2 and x = 2

QUESTION 35

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The vertical asymptote(s) of $f(x) = (x + 1)/(x^2(x - 3))$ is/are

Ax = −1, x = 0, and x = 3
Bx = 3 only
Cx = 0 and x = 3
Dx = 0 only

QUESTION 36

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Let $f(x) = (x^3 - x)/(x^2 - 1)$. Which statement is true?

Af has a horizontal asymptote at y = 0
Bf has a vertical asymptote at x = 1
Cf has a vertical asymptote at x = −1 only
Df has holes at x = ±1 and no vertical asymptotes

QUESTION 37

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 1^+} (x+1)/(x^2-1) =$

A $-\infty$
B $1/2$
C $+\infty$
D 0

QUESTION 38

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to 2^-} x / [(x-2)(x+1)] =$

A $-\infty$
B $+\infty$
C 0
D $2/3$

QUESTION 39

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x \to 0^-} 1/x^3 =$

A $+\infty$
B 0
C 1
D $-\infty$

QUESTION 40

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to\infty} x \cdot \sin(1/x) =$

A∞
B1
CThe limit does not exist
D0

QUESTION 41

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} x^2 \cdot \cos(1/x) =$

A−1
B0
CThe limit does not exist
D1

QUESTION 42

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

If $1 - x^2/2 \le g(x) \le 1 + x^2/2$ for all x near 0, then $\lim_{x\to 0} g(x) =$

A1
BCannot be determined
C0
D2

QUESTION 43

EASY · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x\to\infty} \sin x / (x^2 + 1) =$

AThe limit does not exist
B−1
C0
D1

QUESTION 44

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to\infty} (x + \cos x)/x =$

A0
B1
C−1
D∞

QUESTION 45

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} \sqrt{(x^2 + x^4)}/x =$

AThe limit does not exist
B1
C−1
D0

QUESTION 46

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Let $f(x) = (x^2 - 4)/(x - 2)$ for $x \neq 2$ and $f(2) = k$. For what value of k is f continuous at $x = 2$?

A2
B0
CNo value of k works
D4

QUESTION 47

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Let $f(x) = \{ x^2 - 3x + k$ for $x < 1$; $2x$ for $x \geq 1 \}$. For what value of k is f continuous at $x = 1$?

A2
B4
C3
D6

QUESTION 48

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Let $f(x) = \{ \sin(3x)/x$ for $x \neq 0$; c for $x = 0 \}$. For what value of c is f continuous at $x = 0$?

A3
B0
C1
DThe limit does not exist

QUESTION 49

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Let $f(x) = \{ (x^2 - 1)/(x - 1)$ for $x \neq 1$; 2 for $x = 1 \}$. Which statement is true?

Af is discontinuous at x = 1 because the limit equals 1
B*lim* (x→1) f(x) does not exist
Cf is discontinuous at x = 1 because f(1) is undefined
Df is continuous at x = 1 because *lim* (x→1) f(x) = f(1) = 2

QUESTION 50

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Let $f(x) = (x^2 - 9)/(x^2 - 4x + 3)$. Which describes the discontinuities at $x = 3$ and $x = 1$?

ARemovable at x = 3; infinite at x = 1
BJump at x = 3; infinite at x = 1
CRemovable at both x = 1 and x = 3
DInfinite at x = 3; removable at x = 1

QUESTION 51

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Let $f(x) = \{ x + 1$ for $x < 0$; $x^2 + 1$ for $0 \leq x \leq 1$; $3 - x$ for $x > 1 \}$. Which is true?

Af is discontinuous at x = 1 only
Bf is discontinuous at x = 0 and x = 1
Cf is discontinuous at x = 0 only
Df is continuous for all real x

QUESTION 52

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Let $f(x) = \{ x^2$ for $x < 2$; 5 for $x = 2$; $3x - 4$ for $x > 2 \}$. What type of discontinuity occurs at $x = 2$?

AJump
BNo discontinuity
CRemovable
DInfinite

QUESTION 53

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

Let $f(x) = \sqrt{(x^2 - 9)/(x - 4)}$. On which set is f continuous?

A$(-\infty, -3] \cup [3, \infty)$
B$(-\infty, -3] \cup [3, 4) \cup (4, \infty)$
C$[3, \infty)$
D$[-3, 3]$

QUESTION 54

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

Let $f(x) = x^3 - 4x + 1$. On which interval does the IVT guarantee a value c with $f(c) = 0$?

A$[0, 1]$
B$[2, 3]$
C$[-1, 0]$
D$[-2, -1]$

QUESTION 55

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Which equation must have a solution in the open interval $(0, 1)$?

A $Ax^2 = 5$
B $Be^x = 0$
C $Cx^3 + x = 3$
D $D\cos x = x$

QUESTION 56

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Selected values of a continuous function f are shown.

x	1	2	3	4
f(x)	2	−1	4	3

What is the minimum number of zeros of f on $[1, 4]$?

A 2
B 0
C 1
D 3

QUESTION 57

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The table gives values of f.

x	0.9	0.99	0.999	1.001	1.01	1.1
f(x)	1.9	1.99	1.999	2.001	2.01	2.1

Given that $f(1) = 5$, which statement is true?

A $\lim_{x \to 1} f(x) = 5$ and f is continuous at $x = 1$
B $\lim_{x \to 1} f(x) = 2$ and f is discontinuous at $x = 1$ because $f(1) = 5 \neq 2$
C $\lim_{x \to 1} f(x) = 2$ and f is continuous at $x = 1$
D $\lim_{x \to 1} f(x)$ does not exist

QUESTION 58

MODERATE · MULTIPLE CHOICE ★★★ MANDATORY

If $\lim_{x\to a} f(x) = 2$ and $\lim_{x\to a} g(x) = -3$, then $\lim_{x\to a} [f(x)^2 - 3g(x)]/[f(x) + g(x)] =$

A−13
B−5
C13
D5

QUESTION 59

MODERATE · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $\lim_{x\to 2} [f(x) - 3]/(x - 2) = 4$, then $\lim_{x\to 2} f(x) =$

A4
B3
CCannot be determined
D0

QUESTION 60

HARD · MULTIPLE CHOICE ★★★★★ VERY IMPORTANT

If $\lim_{x\to 0} f(x)/x^2 = 5$, which must be true?

A$\lim_{x\to 0} f(x)/x = 5$
Bf(0) = 5
C$\lim_{x\to 0} f(x) = 5$
D$\lim_{x\to 0} f(x) = 0$

QUESTION 61

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

For what value of k does $\lim_{x\to 4} (2x^2 - 3x - k)/(x - 4)$ exist as a finite number?

A12
B32
C10
D20

QUESTION 62

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

If $\lim_{x\to 1} (x^2 + ax + b)/(x - 1) = 5$, then a =

A1
B3
C−4
D5

QUESTION 63

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} (|x| + x)/x =$

A2
B1
CThe limit does not exist
D0

QUESTION 64

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to 3} (x^2 - 9)/|x - 3| =$

A6
BThe limit does not exist
C0
D−6

QUESTION 65

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x\to 2} (x - 2)/(x^2 - 4) =$

A0
B1/4
CThe limit does not exist
D1/2

QUESTION 66

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

The graph of f satisfies $\lim_{x\to 3^-} f(x) = 5$, $\lim_{x\to 3^+} f(x) = 5$, and $f(3) = 2$. Which is true?

Af is continuous at x = 3
Bf has a removable discontinuity at x = 3
C$\lim_{x\to 3} f(x)$ does not exist
Df has a jump discontinuity at x = 3

QUESTION 67

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

The table gives values of f near $x = 1$.

x	0.9	0.99	0.999	1.001	1.01	1.1
f(x)	−10	−100	−1000	1000	100	10

Which statement is best supported?

A $\lim_{x \to 1} f(x) = 1$
B $\lim_{x \to 1} f(x) = +\infty$
C $\lim_{x \to 1} f(x)$ does not exist; f has a vertical asymptote at $x = 1$
D $\lim_{x \to 1} f(x) = 0$

QUESTION 68

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

A function f is continuous on $[-1, 3]$ with $f(-1) = 4$ and $f(3) = -2$. Which must be true?

Af(2) = 0
BThere is c in (−1, 3) with f(c) = 0
Cf attains its maximum at x = 1
Df is increasing on [−1, 3]

QUESTION 69

HARD · MULTIPLE CHOICE

★★★ MANDATORY

On which set is $f(x) = \sqrt{(x + 2)/(x^2 - 4)}$ continuous?

A$[-2, 2) \cup (2, \infty)$
B$(-\infty, -2) \cup (-2, 2)$
C$[-2, \infty)$
D$(-2, 2) \cup (2, \infty)$

QUESTION 70

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Which function has a removable discontinuity at $x = 0$?

A$|x|/x$
B$1/x$
C$\sin x / x$
D$x/|x|$

QUESTION 71

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x \to 0} (e^{3x} - 1)/(2x) =$

A$1/2$
B3
C$3/2$
D0

QUESTION 72

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x \to \infty} 3^x/(3^x + 1) =$

A0
B1
C∞
D$1/3$

QUESTION 73

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

$\lim_{x\to\infty} (5 - 2e^{-x})/(3 + e^{-x}) =$

A5
B3/5
C∞
D5/3

QUESTION 74

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to-2} (x + 2)/|x + 2| =$

A−1
BThe limit does not exist
C1
D0

QUESTION 75

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Let $\lfloor x \rfloor$ denote the greatest integer function. Then $\lim_{x\to0^+} x\lfloor 1/x \rfloor =$

A1
BThe limit does not exist
C∞
D0

QUESTION 76

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

Let $f(x) = [x]$. Which statement is true at $x = 2$?

A $\lim_{x \to 2^-} f(x) = 2$
B $\lim_{x \to 2} f(x)$ exists and equals 2
C f is continuous at $x = 2$
D $\lim_{x \to 2^-} f(x) = 1$ and $\lim_{x \to 2^+} f(x) = 2$, so $\lim_{x \to 2} f(x)$ does not exist

QUESTION 77

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Let f be continuous on $[0, 4]$ with $f(0) = f(4)$. Which must be true?

AThere is c in [0, 2] with f(c) = f(c + 2)
Bf(c) = f(c + 2) only if f is linear
CThis cannot be decided without the graph
DNot necessarily true; no such c need exist

QUESTION 78

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

Which function is NOT continuous at $x = 0$?

Ae^x
B$\sqrt{(x^2 + 1)}$
C$|x|$
D$x/|x|$

QUESTION 79

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to 0} (1 - \cos(x^2))/x^4 =$

A1
B0
C1/2
D1/4

QUESTION 80

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x\to \infty} x^2 \sin(1/x)/(x + 1) =$

A2
B0
C1
D∞

QUESTION 81

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x\to 1} (x^4 - 1)/(x^3 - 1) =$

A3/4
B1
C4/3
D2

QUESTION 82

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to -1} (x^3 + 1)/(x^2 - 1) =$

A3/2
B−3/2
CThe limit does not exist
D0

QUESTION 83

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x \to 0} (\sqrt{(1 + x)} - 1)/(\sqrt[3]{(1 + x)} - 1) =$

A0
B3/2
C2/3
D1

QUESTION 84

MODERATE · MULTIPLE CHOICE

★★ IMPORTANT

The horizontal asymptote of $f(x) = (2x^2 - 3)/(x^2 + 5)$ is

Ay = −3/5
By = 0
Cy = 5
Dy = 2

QUESTION 85

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

How many horizontal asymptotes does $f(x) = x/(\sqrt{(x^2 + 1)} + x)$ have?

A1
B3
C0
D2

QUESTION 86

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

$\lim_{x \to \infty} (x^2 + 2x)/\sqrt{(x^4 + 1)} =$

A2
B1
C∞
D0

QUESTION 87

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

$\lim_{x \to 0^+} \sqrt{x} \cdot \sin(1/x) =$

A0
BThe limit does not exist
C∞
D1

QUESTION 88

MODERATE · MULTIPLE CHOICE

★★★ MANDATORY

On the open interval $(0, 3)$, where is $f(x) = [x]$ discontinuous?

Af is continuous on (0, 3)
Bx = 1, 2, and 3
Cx = 0, 1, and 2
Dx = 1 and x = 2

QUESTION 89

HARD · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

Let $f(x) = (x^2 + 3x - 10)/(x - 2)$ for $x \neq 2$ and $f(2) = m$. For what m is f continuous at $x = 2$?

A10
B5
C−7
D7

QUESTION 90

MODERATE · MULTIPLE CHOICE

★★★★★ VERY IMPORTANT

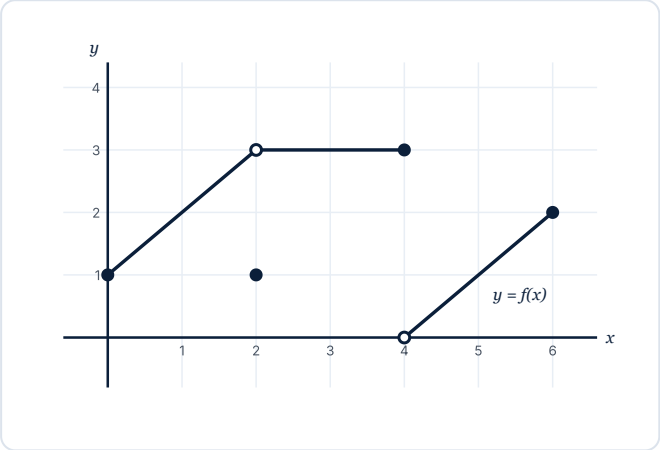
$\lim_{x \to 0} (x^2 + |x|)/x =$

A1
BThe limit does not exist
C0
D−1

Question 91: Analyzing Limits and Continuity from a Graph

★★★★★ VERY IMPORTANT 9 Points

The graph of the function f is shown below for $0 \le x \le 6$. Filled circles indicate included points and open circles indicate excluded points.



Graph of f on the closed interval [0, 6]

(a) [3 points] Find $\lim_{x \to 2^-} f(x)$, $\lim_{x \to 2^+} f(x)$, and $\lim_{x \to 2} f(x)$, or state that a limit does not exist.

(b) [2 points] Is f continuous at x = 2? Justify your answer using the definition of continuity, and classify the discontinuity if one exists.

(c) [2 points] Find $\lim_{x \to 4^-} f(x)$ and $\lim_{x \to 4^+} f(x)$. Classify the discontinuity at x = 4.

(d) [2 points] Write the intervals on which f is continuous.

Reminder: Show the setup of every limit and state any theorem you use by name.

Question 92: Piecewise Function with Two Parameters

★★★★★ VERY IMPORTANT | 9 Points

Let f be the function defined by $\begin{cases} x^2 + a & \text{FOR } x < 1 \\ 3 & \text{FOR } x = 1 \\ bx + 2 & \text{FOR } x > 1 \end{cases}$ where a and b are constants.

(a) [2 points] Find $\lim_{x \to 1^-} f(x)$ and $\lim_{x \to 1^+} f(x)$ in terms of a and b.

(b) [4 points] Determine the values of a and b for which f is continuous at x = 1. Justify your answer using the three conditions of continuity.

(c) [3 points] If a = 2 and b = 5, classify the discontinuity of f at x = 1. Justify your answer.

Reminder: Show the setup of every limit and state any theorem you use by name.

Question 93: Estimating a Limit from a Table of Values

★★★★★ VERY IMPORTANT · 9 Points

The table below gives selected values of a function g.

x	2.9	2.99	2.999	3.001	3.01	3.1
g(x)	6.700	6.970	6.997	7.003	7.030	7.300

It is also given that $g(3) = 4$.

(a) [2 points] Use the table to estimate $\lim_{x \to 3^-} g(x)$ and $\lim_{x \to 3^+} g(x)$.

(b) [2 points] Based on the table, estimate $\lim_{x \to 3} g(x)$ and explain your reasoning.

(c) [3 points] Is g continuous at x = 3? Justify your answer using the definition of continuity.

(d) [2 points] What value should be assigned to g(3) so that g becomes continuous at x = 3? Name the type of discontinuity that is being removed.

Reminder: Show the setup of every limit and state any theorem you use by name.

Question 94: Holes, Vertical Asymptotes, and End Behavior

★★★★★ VERY IMPORTANT · 9 Points

Let h be the function defined by $h(x) = (x^2 - x - 6)/(x^2 - 9)$.

(a) [2 points] Factor the numerator and denominator, and write a simplified expression for h(x), stating any restrictions.

(b) [2 points] Find $\lim_{x \to 3} h(x)$. Explain what this tells you about the graph of h at x = 3.

(c) [3 points] Find $\lim_{x \to -3^-} h(x)$ and $\lim_{x \to -3^+} h(x)$, and give the equation of the vertical asymptote.

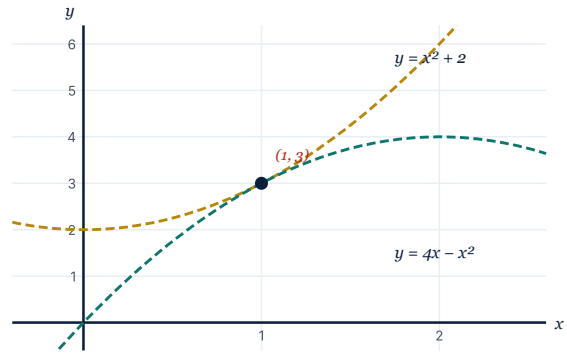
(d) [2 points] Find $\lim_{x \to \infty} h(x)$ and give the equation of the horizontal asymptote.

Reminder: Show the setup of every limit and state any theorem you use by name.

Question 95: The Squeeze Theorem

★★★★★ VERY IMPORTANT | 9 Points

The graph below shows two functions that bound a third function g. For all real x it is given that $4x - x^2 \le g(x) \le x^2 + 2$.



Upper bound $y = x^2 + 2$ (gold) and lower bound $y = 4x - x^2$ (teal)

(a) [2 points] Evaluate ***lim*** (x→1) (4x − x²) and ***lim*** (x→1) (x² + 2).

(b) [3 points] Use the Squeeze Theorem to find ***lim*** (x→1) g(x). State clearly why the theorem applies.

(c) [2 points] Explain why these same bounds do NOT determine ***lim*** (x→0) g(x).

(d) [2 points] Use the Squeeze Theorem to evaluate ***lim*** (x→0) x² cos(1/x).

Reminder: Show the setup of every limit and state any theorem you use by name.

Question 96: The Intermediate Value Theorem with Tabular Data

★★★★★ VERY IMPORTANT 9 Points

The function f is continuous on the closed interval [0, 8]. Selected values of f are given in the table.

x	0	2	4	6	8
f(x)	5	−2	3	−1	4

(a) [2 points] Explain why the Intermediate Value Theorem may be applied to f on [0, 8].

(b) [4 points] What is the minimum number of values of x in (0, 8) for which f(x) = 0? Justify your answer.

(c) [2 points] Must there be a value c in (0, 2) such that f(c) = 1? Justify your answer.

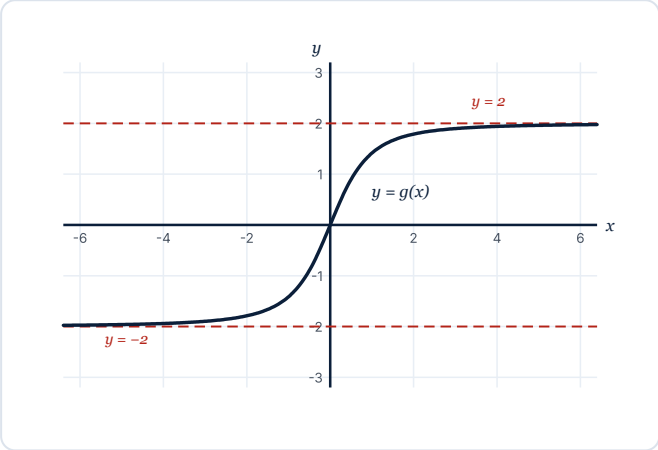
(d) [1 points] Could f have more zeros on [0, 8] than the minimum found in part (b)? Explain.

Reminder: Show the setup of every limit and state any theorem you use by name.

Question 97: Limits at Infinity and Horizontal Asymptotes

★★★★★ VERY IMPORTANT 9 Points

The graph of $g(x) = 2x / \sqrt{(x^2 + 1)}$ is shown below together with its horizontal asymptotes, drawn as dashed lines.



Graph of $g(x) = 2x / \sqrt{(x^2 + 1)}$ with its two horizontal asymptotes (dashed)

(a) [3 points] Use the graph to state ***lim*** $(x \to \infty)$ $g(x)$, then verify this value algebraically.

(b) [4 points] Use the graph to state ***lim*** $(x \to -\infty)$ $g(x)$, then verify this value algebraically. Explain carefully why the answer differs in sign from part (a).

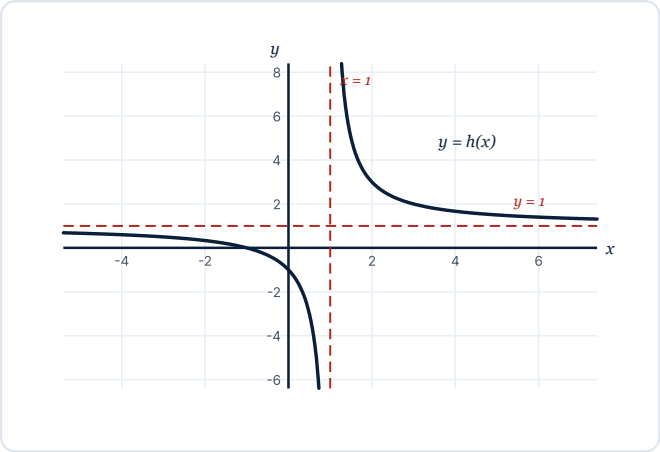
(c) [2 points] Does the graph of g have any vertical asymptotes? Justify your answer.

Reminder: Show the setup of every limit and state any theorem you use by name.

Question 98: Infinite Limits and Asymptotic Behavior from a Graph

★★★★★ VERY IMPORTANT 9 Points

The graph of $h(x) = (x + 1)/(x - 1)$ is shown below. The dashed lines indicate asymptotes.



Graph of $h(x) = (x + 1)/(x - 1)$

(a) [2 points] Use the graph to find $\lim_{x \to 1^-} h(x)$ and $\lim_{x \to 1^+} h(x)$.

(b) [3 points] Verify your answers to part (a) algebraically by analyzing the sign of the numerator and denominator.

(c) [2 points] State the equation of the vertical asymptote and classify the discontinuity at that location.

(d) [2 points] Find $\lim_{x \to \infty} h(x)$ and $\lim_{x \to -\infty} h(x)$, and state the equation of the horizontal asymptote.

Reminder: Show the setup of every limit and state any theorem you use by name.

Question 99: Special Trigonometric Limits and Continuity

★★★★★ VERY IMPORTANT | 9 Points

Let f be the function defined by $\begin{cases} \sin(5x)/(3x) & \text{FOR } x < 0 \\ c & \text{FOR } x = 0 \\ (1 - \cos 2x)/x^2 & \text{FOR } x > 0 \end{cases}$ where c is a constant.

(a) [3 points] Evaluate ***lim*** $(x \to 0^-)$ f(x). Show the algebraic steps that lead to your answer.

(b) [4 points] Evaluate ***lim*** $(x \to 0^+)$ f(x). Show the algebraic steps that lead to your answer.

(c) [2 points] Determine all values of c, if any, for which f is continuous at x = 0. Justify your answer.

Reminder: Show the setup of every limit and state any theorem you use by name.

Question 100: Applied Context: Cooling Model

★★★★★ VERY IMPORTANT · 9 Points

A cup of coffee is placed in a room. Its temperature T, in degrees Celsius, after t minutes is modeled by $T(t) = 18 + 72e^{(-0.04t)}$ for $t \geq 0$. Selected values of T are given in the table, rounded to the nearest tenth.

t (min)	0	10	20	40	60
T(t) (°C)	90.0	66.3	50.4	32.5	24.5

(a) [2 points] Find the initial temperature T(0) and verify that it is consistent with the table.

(b) [2 points] Using the table, find the average rate of change of T on the interval [0, 20]. Include units.

(c) [3 points] Evaluate $\lim_{t \to \infty} T(t)$ and interpret the meaning of this limit in the context of the problem.

(d) [2 points] Explain why T is continuous on [0, 60], and then use the Intermediate Value Theorem to show that the coffee reaches exactly 40°C at some time t in (0, 60).

Reminder: Show the setup of every limit and state any theorem you use by name.

End of Question Paper

You have reached the end of Questions 1 through 100.

Before you continue: Make sure you have genuinely attempted every question. The pages that follow contain the complete answer key, fully worked multiple choice solutions with analysis of every incorrect option, and model free response solutions with AP-style scoring rubrics.

What Comes Next

- **Answer Key:** Letter answers for Questions 1–90 with priority ratings.
- **Multiple Choice Solutions:** Step-by-step reasoning for Questions 1–90 plus an explanation of why each distractor fails.
- **Free Response Solutions:** Complete model answers for Questions 91–100 with point-by-point rubrics.
- **Appendix:** Essential Unit 1 formulas and theorems.

Multiple Choice Answer Key

Letter answers with priority star ratings. Questions 91–100 are free response and are scored using the rubrics that follow.

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
B ★★★	D ★★	C ★★★	B ★★★	B ★★★	A ★★★	D ★★★	B ★★★	D ★★★★★	B ★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
C ★★★	D ★★★★★	B ★★★★★	C ★★★★★	C ★★★★★	A ★★★	D ★★★	C ★★★★★	B ★★★	D ★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
B ★★★	D ★★	A ★★★	C ★★★★★	C ★★★	A ★★	D ★★★	A ★★	B ★★★	A ★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
B ★★★★★	A ★★★	B ★★★	B ★★★★★	C ★★★	D ★★★★★	C ★★★	A ★★★	D ★★	B ★★★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
B ★★★★★	A ★★★	C ★★	B ★★★	A ★★★★★	D ★★★	B ★★★★★	A ★★★★★	D ★★★	A ★★★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
D ★★★	A ★★★	B ★★★	A ★★★★★	D ★★★	A ★★★★★	B ★★★	A ★★★	B ★★★★★	D ★★★★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
D ★★★	B ★★★★★	C ★★★★★	B ★★★★★	B ★★	B ★★★★★	C ★★★	B ★★★	D ★★★	C ★★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
C ★★	B ★★	D ★★	B ★★★	A ★★★★★	D ★★★	A ★★★★★	D ★★	C ★★★★★	C ★★★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
C ★★★	B ★★★	B ★★★★★	D ★★	A ★★★★★	B ★★★	A ★★★★★	D ★★★	D ★★★★★	B ★★★★★

Scoring Guide: Each multiple choice question is worth 1 point (90 total). Each free response question is worth 9 points (90 total). Maximum raw score: 180 points.

Question 01

★★★ MANDATORY

Correct Answer: (B) 4

STEP-BY-STEP SOLUTION

Factor the numerator: $x^2 - 4 = (x - 2)(x + 2)$. Cancel $(x - 2)$ to obtain $x + 2$. Then $\lim_{x \to 2}(x + 2) = 4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Substitutes $x = 2$ into the numerator only.

✗ **(C)** Confuses the x-value 2 with the limit value.

✗ **(D)** A 0/0 form does not mean the limit fails; factoring resolves it.

Question 02

★★ IMPORTANT

Correct Answer: (D) −6

STEP-BY-STEP SOLUTION

Polynomials are continuous, so substitute directly: $2(1) + 3(-1) - 5 = 2 - 3 - 5 = -6$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Sign error on the constant term.

✗ **(B)** Sign error on the x^2 term.

✗ **(C)** Sign error on the $3x$ term.

Question 03

★★★ MANDATORY

Correct Answer: (C) 2/3

STEP-BY-STEP SOLUTION

Factor: numerator $= (x - 3)(x + 1)$, denominator $= (x - 3)(x + 3)$. Cancel to get $(x + 1)/(x + 3) \to 4/6 = 2/3$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Substitutes into the numerator only.

✗ **(B)** Factoring removes the 0/0 form, so the limit exists.

✗ **(D)** Wrongly assumes the quotient simplifies to 1.

Question 04

★★★ MANDATORY

Correct Answer: (B) 12

STEP-BY-STEP SOLUTION

Sum of cubes: $x^3 + 8 = (x + 2)(x^2 - 2x + 4)$. Cancel $(x + 2)$: $x^2 - 2x + 4$. At $x = -2$: $4 + 4 + 4 = 12$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses only the x^2 term.

✗ **(C)** Drops the $-2x$ term.

✗ **(D)** Substitutes into the numerator only.

Question 05

★★★ MANDATORY

Correct Answer: (B) 4

STEP-BY-STEP SOLUTION

Write $x - 4 = (\sqrt{x} - 2)(\sqrt{x} + 2)$. Cancelling gives $\sqrt{x} + 2$, which equals 4 at $x = 4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Substitutes into the numerator only.

✗ **(C)** Doubles the correct value.

✗ **(D)** Uses $\sqrt{4} = 2$ and forgets the +2.

Question 06

★★★ MANDATORY

Correct Answer: (A) 1/6

STEP-BY-STEP SOLUTION

Multiply by the conjugate. The numerator becomes x, leaving $1/(\sqrt{(9 + x)} + 3) \rightarrow 1/6$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Substitutes into the numerator only.

✗ **(C)** Uses $1/\sqrt{9}$ and skips the conjugate step.

✗ **(D)** Uses an invalid denominator of 2.

Question 07

★★★ MANDATORY

Correct Answer: (D) −1/4

STEP-BY-STEP SOLUTION

Combine: $1/(x+1) - 1/2 = (1 - x)/(2(x+1))$. Dividing by $(x - 1)$ gives $-1/(2(x+1)) \to -1/4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Drops the factor 2 in the denominator.

✗ **(B)** Sign error treating $1 - x$ as $x - 1$.

✗ **(C)** Combines both errors.

Question 08

★★★ MANDATORY

Correct Answer: (B) −2

STEP-BY-STEP SOLUTION

Factor x from both: $x(x + 4) / [x(x^2 - 2)] = (x + 4)/(x^2 - 2) \to 4/(-2) = -2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Substitutes into the numerator only.

✗ **(C)** Sign error in the denominator.

✗ **(D)** Factoring x removes the 0/0 form.

Question 09

★★★★★ VERY IMPORTANT

Correct Answer: (D) −1/2

STEP-BY-STEP SOLUTION

Rationalize: $[(1 + 2x) - (1 + x)^2] / [x^2(\sqrt{(1+2x)} + 1 + x)] = -x^2 / [x^2(\ldots)] \to -1/2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Stops before dividing by x^2.

✗ **(B)** Sign error expanding $(1 + x)^2$.

✗ **(C)** Rationalization resolves the indeterminate form.

Question 10

★★★ MANDATORY

Correct Answer: (B) 3/4

STEP-BY-STEP SOLUTION

Factor by grouping: $x^3 - 2x^2 - x + 2 = (x - 2)(x^2 - 1)$. Cancel $(x - 2)$: $(x^2 - 1)/(x + 2) \rightarrow 3/4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Substitutes into the numerator only.

✗ **(C)** Incorrect factorization.

✗ **(D)** Forgets to divide by $(x + 2)$.

Question 11

★★★ MANDATORY

Correct Answer: (C) 7/6

STEP-BY-STEP SOLUTION

Factor: $(2x - 1)(x + 3) / [(x - 3)(x + 3)] = (2x - 1)/(x - 3) \rightarrow (-7)/(-6) = 7/6$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Cancelling removes the 0/0 form.

✗ **(B)** Sign error evaluating $x - 3$ at $x = -3$.

✗ **(D)** Incorrect factorization.

Question 12

★★★★★ VERY IMPORTANT

Correct Answer: (D) The limit does not exist

STEP-BY-STEP SOLUTION

Left-hand limit: $2^2 - 1 = 3$. Right-hand limit: $3(2) - 2 = 4$. Since $3 \neq 4$, the two-sided limit does not exist.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses only the left-hand piece.

✗ **(B)** 2 is the x-value, not a limit value.

✗ **(C)** Uses only the right-hand piece.

Question 13

★★★★★ VERY IMPORTANT

Correct Answer: (B) 2

STEP-BY-STEP SOLUTION

Left value: 2 + k. Right value: 1 + 3 = 4. Continuity requires 2 + k = 4, so k = 2.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Sets 2 + k = 2.

✗ **(C)** Sets k equal to the right-hand value 4.

✗ **(D)** Uses a miscalculated right-hand value of 3.

Question 14

★★★★★ VERY IMPORTANT

Correct Answer: (C) 3/2

STEP-BY-STEP SOLUTION

Write as [sin 3x / x] · [sin 3x / sin 6x]. The first factor → 3; the second → 3/6 = 1/2. Product = 3/2.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Drops the factor 3.

✗ **(B)** Substitutes sin 0 = 0 without handling the denominator.

✗ **(D)** Ignores the factor 1/2.

Question 15

★★★★★ VERY IMPORTANT

Correct Answer: (C) 9/2

STEP-BY-STEP SOLUTION

Use $(1 - \cos u)/u^2 \to 1/2$ with $u = 3x$. Then the expression equals $9 \cdot (1 - \cos 3x)/(3x)^2 \to 9/2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Ignores the x^2 denominator.

✗ **(B)** Forgets to scale by 3^2.

✗ **(D)** Scales twice by the coefficient.

Question 16

★★★ MANDATORY

Correct Answer: (A) 1/5

STEP-BY-STEP SOLUTION

$x/\tan 5x = (1/5)\cdot[5x/\sin 5x]\cdot\cos 5x \to (1/5)(1)(1) = 1/5$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Ignores the coefficient 5.

✗ **(C)** Inverts the ratio.

✗ **(D)** Treats the limit as 0 from the numerator.

Question 17

★★★ MANDATORY

Correct Answer: (D) 2

STEP-BY-STEP SOLUTION

Divide numerator and denominator by x: $[\sin 4x / x] / [\sin 2x / x] \to 4/2 = 2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses only the numerator coefficient.

✗ **(B)** Inverts the ratio.

✗ **(C)** Assumes both sine ratios cancel to 1.

Question 18

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1/2

STEP-BY-STEP SOLUTION

$\tan x - \sin x = \sin x(1 - \cos x)/\cos x$. So the expression $= (\sin x / x)\cdot[(1 - \cos x)/x^2]\cdot(1/\cos x) \to 1\cdot(1/2)\cdot 1 = 1/2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Assumes the numerator forces the limit to 0.

✗ **(B)** Treats it as a $\sin x / x$ limit.

✗ **(D)** Uses only the cubic term of $\tan x$.

Question 19

★★★ MANDATORY

Correct Answer: (B) −1

STEP-BY-STEP SOLUTION

Let $h = x - \pi$. Then $\sin(\pi + h) = -\sin h$, so the expression $= -\sin h / h \to -1$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses $\sin \pi = 0$ in the numerator only.

✗ **(C)** The substitution $h = x - \pi$ gives a finite limit.

✗ **(D)** Ignores the sign change of the shifted sine.

Question 20

★★★ MANDATORY

Correct Answer: (D) 1

STEP-BY-STEP SOLUTION

Since $\cos x = \sin(\pi/2 - x)$, let $u = \pi/2 - x \to 0$. The expression becomes $\sin u / u \to 1$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses $\cos(\pi/2) = 0$ only.

✗ **(B)** Introduces an incorrect sign.

✗ **(C)** Confuses the x-value with the limit.

Question 21

★★★ MANDATORY

Correct Answer: (B) 1/2

STEP-BY-STEP SOLUTION

Write as $[(1 - \cos x)/x^2] \cdot [x/\sin x] \to (1/2)(1) = 1/2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Both factors are finite, so the limit exists.

✗ **(C)** Confuses this with $\sin x / x$.

✗ **(D)** Ignores the $\sin x$ factor in the denominator.

Question 22

★★ IMPORTANT

Correct Answer: (D) 0

STEP-BY-STEP SOLUTION

Write $\sin(x^2)/x = x \cdot [\sin(x^2)/x^2] \to 0 \cdot 1 = 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Forgets the leftover factor x.

✗ **(B)** The quotient stays bounded near 0.

✗ **(C)** The numerator vanishes faster than the denominator.

Question 23

★★★ MANDATORY

Correct Answer: (A) 1

STEP-BY-STEP SOLUTION

Since $1 - \cos^2 x = \sin^2 x$, the expression is $(\sin x / x)^2 \to 1$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Mistakes the identity for $2\sin^2 x$.

✗ **(C)** The identity makes the limit finite.

✗ **(D)** Substitutes into the numerator only.

Question 24

★★★★★ VERY IMPORTANT

Correct Answer: (C) 0

STEP-BY-STEP SOLUTION

Since $-1 \le \sin(1/x) \le 1$, we get $-|x| \le |x| \sin(1/x) \le |x|$. Both bounds $\to 0$, so by the Squeeze Theorem the limit is 0.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The $|x|$ factor damps the oscillation.

✗ **(B)** Assumes $\sin(1/x) \to 1$.

✗ **(D)** Uses the lower bound of the sine.

Question 25

★★★ MANDATORY

Correct Answer: (C) 1/2

STEP-BY-STEP SOLUTION

$\sec x - 1 = (1 - \cos x)/\cos x$, so the expression = $[(1 - \cos x)/x^2]\cdot(1/\cos x) \to 1/2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses the wrong special limit.

✗ **(B)** Substitutes directly into the numerator.

✗ **(D)** The limit at 0 is finite.

Question 26

★★ IMPORTANT

Correct Answer: (A) 4

STEP-BY-STEP SOLUTION

Equal degrees, so the limit is the ratio of leading coefficients: 4/1 = 4.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Assumes the denominator dominates.

✗ **(C)** Inverts the ratio.

✗ **(D)** Equal degrees give a finite limit.

Question 27

★★★ MANDATORY

Correct Answer: (D) 3/2

STEP-BY-STEP SOLUTION

Equal degrees (3), so the limit is 3/2.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Ignores the leading coefficient 2.

✗ **(B)** Assumes the denominator dominates.

✗ **(C)** Equal degrees give a finite limit.

Question 28

★★ IMPORTANT

Correct Answer: (A) ∞

STEP-BY-STEP SOLUTION

Numerator degree 2 exceeds denominator degree 1, and both are positive for large x, so the limit is $+\infty$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Leading-coefficient ratio requires equal degrees.

✗ **(C)** Assumes the denominator dominates.

✗ **(D)** The quotient is positive for large positive x.

Question 29

★★★ MANDATORY

Correct Answer: (B) −∞

STEP-BY-STEP SOLUTION

The numerator is positive while $2x + 3$ is large and negative, so the quotient tends to $-\infty$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Leading-coefficient ratio requires equal degrees.

✗ **(C)** Sign error on the denominator.

✗ **(D)** Assumes the denominator dominates.

Question 30

★★★ MANDATORY

Correct Answer: (A) 2

STEP-BY-STEP SOLUTION

For $x > 0$, $\sqrt{(x^2 + 1)} = x\sqrt{(1 + 1/x^2)}$, so the quotient $\to 2$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Drops the coefficient 2.

✗ **(C)** Both grow at the same rate.

✗ **(D)** Assumes the denominator dominates.

Question 31

★★★★★ VERY IMPORTANT

Correct Answer: (B) −2

STEP-BY-STEP SOLUTION

For $x < 0$, $\sqrt{(x^2 + 1)} = -x\sqrt{(1 + 1/x^2)}$, so the quotient $\to -2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses $\sqrt{(x^2)} = x$, which fails for $x < 0$.

✗ **(C)** The limit at $-\infty$ exists.

✗ **(D)** Both grow at the same rate.

Question 32

★★★ MANDATORY

Correct Answer: (A) 1/4

STEP-BY-STEP SOLUTION

Multiply by the conjugate: $x / [x(\sqrt{(4 + 1/x)} + 2)] \to 1/4$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Forgets the +2x term in the denominator.

✗ **(C)** $\infty - \infty$ is not automatically 0.

✗ **(D)** Uses only the constant 2.

Question 33

★★★ MANDATORY

Correct Answer: (B) 3/2

STEP-BY-STEP SOLUTION

Conjugate gives $3x / [x(\sqrt{(1 + 3/x)} + 1)] \to 3/2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Drops the x factored out of the denominator.

✗ **(C)** The difference converges.

✗ **(D)** Assumes the terms cancel completely.

Question 34

★★★★★ VERY IMPORTANT

Correct Answer: (B) x = −1 only

STEP-BY-STEP SOLUTION

Factor: $(x - 2)(x + 2) / [(x - 2)(x + 1)] = (x + 2)/(x + 1)$. Only $x = -1$ remains; $x = 2$ is a removable hole.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** $x = 2$ is a hole.

✗ **(C)** $x = 2$ is a hole, not an asymptote.

✗ **(D)** $x = -2$ is an x-intercept.

Question 35

★★★ MANDATORY

Correct Answer: (C) x = 0 and x = 3

STEP-BY-STEP SOLUTION

The denominator vanishes at $x = 0$ and $x = 3$, and the numerator is nonzero at both, so both are vertical asymptotes.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** $x = -1$ makes the numerator zero.

✗ **(B)** Misses $x = 0$.

✗ **(D)** Misses $x = 3$.

Question 36

★★★★★ VERY IMPORTANT

Correct Answer: (D) f has holes at x = ±1 and no vertical asymptotes

STEP-BY-STEP SOLUTION

Factor: $x(x - 1)(x + 1) / [(x - 1)(x + 1)] = x$ for $x \neq \pm 1$. Both factors cancel, so $x = \pm 1$ are holes.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** $f(x) = x$ grows without bound.

✗ **(B)** $x = 1$ is a hole.

✗ **(C)** $x = -1$ is also a hole.

Question 37

★★★ MANDATORY

Correct Answer: (C) +∞

STEP-BY-STEP SOLUTION

Simplify to 1/(x − 1). From the right, x − 1 is a small positive number, so the value → +∞.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is the left-hand behavior.

✗ **(B)** Substitutes while ignoring the zero denominator.

✗ **(D)** The denominator vanishes.

Question 38

★★★ MANDATORY

Correct Answer: (A) −∞

STEP-BY-STEP SOLUTION

Near x = 2 the function behaves like 2 / [3(x − 2)]. From the left, x − 2 is small and negative, so the value → −∞.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Sign error from the left side.

✗ **(C)** The numerator is nonzero at 2.

✗ **(D)** Ignores the vanishing factor.

Question 39

★★ IMPORTANT

Correct Answer: (D) −∞

STEP-BY-STEP SOLUTION

For $x < 0$, x^3 is a small negative number, so the reciprocal → −∞.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Sign error for negative x.

✗ **(B)** The denominator shrinks rather than grows.

✗ **(C)** Treats x^3 as 1.

Question 40

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1

STEP-BY-STEP SOLUTION

Let $u = 1/x \to 0^+$. Then the expression becomes $\sin u / u \to 1$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The factors cancel rather than grow.

✗ **(C)** $1/x \to 0$, so no oscillation occurs.

✗ **(D)** The product behaves like $x \cdot (1/x)$.

Question 41

★★★★★ VERY IMPORTANT

Correct Answer: (B) 0

STEP-BY-STEP SOLUTION

Since $-x^2 \le x^2 \cos(1/x) \le x^2$ and both bounds $\to 0$, the Squeeze Theorem gives 0.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses the lower bound of the cosine.

✗ **(C)** The x^2 factor forces convergence.

✗ **(D)** Assumes $\cos(1/x) \to 1$.

Question 42

★★★ MANDATORY

Correct Answer: (A) 1

STEP-BY-STEP SOLUTION

Both bounding functions approach 1, so by the Squeeze Theorem $\lim_{x \to 0} g(x) = 1$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The shared limit makes the conclusion valid.

✗ **(C)** The bounds approach 1, not 0.

✗ **(D)** Adds the two bound values.

Question 43

★★ IMPORTANT

Correct Answer: (C) 0

STEP-BY-STEP SOLUTION

Since $-1 \le \sin x \le 1$, the quotient is squeezed between $\pm 1/(x^2 + 1)$, both of which $\to 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The growing denominator damps the oscillation.

✗ **(B)** Uses the lower bound of the sine.

✗ **(D)** sin x is not constantly 1.

Question 44

★★★ MANDATORY

Correct Answer: (B) 1

STEP-BY-STEP SOLUTION

Split: $1 + (\cos x)/x$. Since $|\cos x / x| \le 1/x \to 0$, the limit is 1.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The leading term dominates and is nonzero.

✗ **(C)** The constant term is +1.

✗ **(D)** cos x is bounded, so no growth occurs.

Question 45

★★★★★ VERY IMPORTANT

Correct Answer: (A) The limit does not exist

STEP-BY-STEP SOLUTION

$\sqrt{(x^2 + x^4)} = |x|\sqrt{(1 + x^2)}$. From the right the quotient $\to 1$; from the left it $\to -1$, so the limit does not exist.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** This is only the right-hand limit.

✗ **(C)** This is only the left-hand limit.

✗ **(D)** The expression is not identically zero.

Question 46

★★★ MANDATORY

Correct Answer: (D) 4

STEP-BY-STEP SOLUTION

For $x \neq 2$, $f(x) = x + 2$, so the limit at 2 is 4. Continuity requires $k = 4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Confuses the x-value with the limit.

✗ **(B)** Uses the numerator value at $x = 2$.

✗ **(C)** The limit exists, so a value does work.

Question 47

★★★★★ VERY IMPORTANT

Correct Answer: (B) 4

STEP-BY-STEP SOLUTION

Left limit: $1 - 3 + k = k - 2$. Right value: 2. Setting $k - 2 = 2$ gives $k = 4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Sets k equal to the right-hand value.

✗ **(C)** Miscomputes $1 - 3$ as -1.

✗ **(D)** Uses the wrong right-hand value.

Question 48

★★★★★ VERY IMPORTANT

Correct Answer: (A) 3

STEP-BY-STEP SOLUTION

$\lim_{x \to 0} \sin(3x)/x = 3$, so continuity requires $c = 3$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Uses $\sin 0 = 0$.

✗ **(C)** Forgets the factor 3.

✗ **(D)** The special limit is finite.

Question 49

★★★ MANDATORY

Correct Answer: (D) f is continuous at x = 1 because *lim* (x→1) f(x) = f(1) = 2

STEP-BY-STEP SOLUTION

For $x \neq 1$ the expression simplifies to $x + 1$, whose limit is 2, and $f(1) = 2$. All three continuity conditions hold.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The limit is 2, not 1.

✗ **(B)** The limit exists and equals 2.

✗ **(C)** f(1) is defined as 2.

Question 50

★★★★★ VERY IMPORTANT

Correct Answer: (A) Removable at x = 3; infinite at x = 1

STEP-BY-STEP SOLUTION

Factoring gives $(x - 3)(x + 3)/[(x - 1)(x - 3)] = (x + 3)/(x - 1)$. At $x = 3$ the limit is 3 (removable); at $x = 1$ the function is unbounded.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The one-sided limits at 3 agree.

✗ **(C)** At x = 1 the limit is infinite.

✗ **(D)** The two cases are reversed.

Question 51

★★★ MANDATORY

Correct Answer: (D) f is continuous for all real x

STEP-BY-STEP SOLUTION

At $x = 0$ all three values equal 1; at $x = 1$ all three values equal 2. Therefore f is continuous everywhere.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The values all agree at x = 1.

✗ **(B)** Both junctions are continuous.

✗ **(C)** The values all agree at x = 0.

Question 52

★★★ MANDATORY

Correct Answer: (A) Jump

STEP-BY-STEP SOLUTION

Left limit = 4, right limit = 2. Both are finite but unequal, which is a jump discontinuity.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The one-sided limits differ.

✗ **(C)** A removable discontinuity requires the two-sided limit to exist.

✗ **(D)** Neither one-sided limit is infinite.

Question 53

★★★ MANDATORY

Correct Answer: (B) $(-\infty, -3] \cup [3, 4) \cup (4, \infty)$

STEP-BY-STEP SOLUTION

We need $x^2 - 9 \geq 0$ (so $x \leq -3$ or $x \geq 3$) and $x \neq 4$. f is continuous on its entire domain.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Fails to exclude $x = 4$.

✗ **(C)** Omits the branch $x \leq -3$.

✗ **(D)** Uses the reversed inequality.

Question 54

★★★★★ VERY IMPORTANT

Correct Answer: (A) [0, 1]

STEP-BY-STEP SOLUTION

$f(0) = 1$ and $f(1) = -2$ have opposite signs, so the IVT guarantees a zero in $(0, 1)$.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** $f(2) = 1$ and $f(3) = 16$ share a sign.

✗ **(C)** $f(-1) = 4$ and $f(0) = 1$ share a sign.

✗ **(D)** $f(-2) = 1$ and $f(-1) = 4$ share a sign.

Question 55

★★★ MANDATORY

Correct Answer: (D) cos x = x

STEP-BY-STEP SOLUTION

Let f(x) = cos x − x. Then f(0) = 1 > 0 and f(1) ≈ −0.46 < 0, so the IVT gives a root in (0, 1).

WHY THE OTHER OPTIONS FAIL

✗ **(A)** $x^2 - 5$ is negative throughout (0, 1).

✗ **(B)** e^x is always positive.

✗ **(C)** $x^3 + x - 3$ is negative at both endpoints.

Question 56

★★★★★ VERY IMPORTANT

Correct Answer: (A) 2

STEP-BY-STEP SOLUTION

Sign changes occur on (1, 2) and on (2, 3); there is no sign change on (3, 4). Therefore at least two zeros are guaranteed.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Ignores both sign changes.

✗ **(C)** Counts only one sign change.

✗ **(D)** No third sign change is guaranteed.

Question 57

★★★ MANDATORY

Correct Answer: (B) $\lim_{x \to 1} f(x) = 2$ and f is discontinuous at x = 1 because f(1) = 5 ≠ 2

STEP-BY-STEP SOLUTION

The table values approach 2 from both sides, so the limit is 2. Since f(1) = 5 ≠ 2, continuity fails.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The table values approach 2.

✗ **(C)** f(1) = 5, so f is not continuous.

✗ **(D)** Both one-sided limits agree.

Question 58

★★★ MANDATORY

Correct Answer: (A) −13

STEP-BY-STEP SOLUTION

Numerator → 4 + 9 = 13; denominator → 2 − 3 = −1. The quotient is −13.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** Numerator computed as 4 − 9.

✗ **(C)** Forgets to divide.

✗ **(D)** Combines a numerator error with a sign error.

Question 59

★★★★★ VERY IMPORTANT

Correct Answer: (B) 3

STEP-BY-STEP SOLUTION

The denominator → 0 while the quotient is finite, so the numerator must → 0. Hence $\lim f(x) = 3$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Confuses the quotient limit with lim f(x).

✗ **(C)** The value is determined.

✗ **(D)** Confuses the numerator limit with lim f(x).

Question 60

★★★★★ VERY IMPORTANT

Correct Answer: (D) $\lim_{x\to 0} f(x) = 0$

STEP-BY-STEP SOLUTION

$f(x) = x^2 \cdot [f(x)/x^2] \to 0 \cdot 5 = 0$. Also $f(x)/x = x \cdot [f(x)/x^2] \to 0$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This quotient tends to 0, not 5.

✗ **(B)** The limit says nothing about the value at 0.

✗ **(C)** The limit must be 0.

Question 61

★★★ MANDATORY

Correct Answer: (D) 20

STEP-BY-STEP SOLUTION

The numerator must vanish at $x = 4$: $32 - 12 - k = 0$, so $k = 20$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses only the $-3x$ term.

✗ **(B)** Uses only the $2x^2$ term.

✗ **(C)** Arithmetic slip combining 32 and 12.

Question 62

★★★★★ VERY IMPORTANT

Correct Answer: (B) 3

STEP-BY-STEP SOLUTION

Write $x^2 + ax + b = (x - 1)(x + c)$. The limit equals $1 + c = 5$, so $c = 4$, giving $a = c - 1 = 3$ and $b = -4$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Guesses from the pattern without solving.

✗ **(C)** This is the value of b.

✗ **(D)** Confuses a with the limit value.

Question 63

★★★★★ VERY IMPORTANT

Correct Answer: (C) The limit does not exist

STEP-BY-STEP SOLUTION

For $x > 0$ the quotient is 2; for $x < 0$ it is 0. The one-sided limits differ, so the limit does not exist.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is only the right-hand limit.

✗ **(B)** Averages the one-sided limits.

✗ **(D)** This is only the left-hand limit.

Question 64

★★★★★ VERY IMPORTANT

Correct Answer: (B) The limit does not exist

STEP-BY-STEP SOLUTION

From the right the quotient → 6; from the left it → −6. The one-sided limits differ.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is only the right-hand limit.

✗ **(C)** Substitutes into the numerator only.

✗ **(D)** This is only the left-hand limit.

Question 65

★★ IMPORTANT

Correct Answer: (B) 1/4

STEP-BY-STEP SOLUTION

Cancel $(x - 2)$ to get $1/(x + 2)$, which equals $1/4$ at $x = 2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Substitutes into the numerator only.

✗ **(C)** Factoring resolves the indeterminate form.

✗ **(D)** Forgets the factor $(x + 2)$.

Question 66

★★★★★ VERY IMPORTANT

Correct Answer: (B) f has a removable discontinuity at x = 3

STEP-BY-STEP SOLUTION

Both one-sided limits equal 5, so the limit exists. Since $f(3) = 2 \neq 5$, the discontinuity is removable.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Continuity requires $f(3) = 5$.

✗ **(C)** The one-sided limits agree.

✗ **(D)** A jump requires unequal one-sided limits.

Question 67

★★★ MANDATORY

Correct Answer: (C) $\lim_{x\to 1} f(x)$ does not exist; f has a vertical asymptote at x = 1

STEP-BY-STEP SOLUTION

From the left the values decrease without bound; from the right they increase without bound. The behaviors differ, so the limit does not exist and x = 1 is a vertical asymptote.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The values do not approach 1.
✗ **(B)** This ignores the left-hand behavior.
✗ **(D)** The values are not approaching 0.

Question 68

★★★ MANDATORY

Correct Answer: (B) There is c in (−1, 3) with f(c) = 0

STEP-BY-STEP SOLUTION

Since 0 lies between 4 and −2 and f is continuous, the IVT guarantees some c in (−1, 3) with f(c) = 0.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The IVT does not identify the location of the zero.
✗ **(C)** The IVT says nothing about where extrema occur.
✗ **(D)** No monotonicity information is given.

Question 69

★★★ MANDATORY

Correct Answer: (D) $(-2, 2) \cup (2, \infty)$

STEP-BY-STEP SOLUTION

The radical requires $x \geq -2$ and the denominator requires $x \neq \pm 2$. Since $x = -2$ makes the denominator zero, it must also be excluded.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Includes $x = -2$ where the denominator vanishes.
✗ **(B)** Uses the wrong domain for the radical.
✗ **(C)** Ignores the zeros of the denominator.

Question 70

★★★ MANDATORY

Correct Answer: (C) sin x / x

STEP-BY-STEP SOLUTION

sin x / x is undefined at 0 but has limit 1, so defining the value 1 removes the discontinuity.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This has a jump discontinuity.

✗ **(B)** 1/x has an infinite discontinuity.

✗ **(D)** This also has a jump discontinuity.

Question 71

★★ IMPORTANT

Correct Answer: (C) 3/2

STEP-BY-STEP SOLUTION

Since (e^(3x) − 1)/(3x) → 1, the expression equals (3/2)·[(e^(3x) − 1)/(3x)] → 3/2.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Forgets the factor 3.

✗ **(B)** Ignores the denominator 2.

✗ **(D)** Substitutes into the numerator only.

Question 72

★★ IMPORTANT

Correct Answer: (B) 1

STEP-BY-STEP SOLUTION

Divide numerator and denominator by 3^x to get $1/(1 + 3^{-x}) \to 1$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The numerator does not vanish.

✗ **(C)** Both terms grow at the same rate.

✗ **(D)** Uses an incorrect ratio.

Question 73

★★ IMPORTANT

Correct Answer: (D) 5/3

STEP-BY-STEP SOLUTION

As x → ∞, e^(−x) → 0, so the expression → 5/3.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Ignores the constant 3.

✗ **(B)** Inverts the ratio.

✗ **(C)** The exponential terms decay.

Question 74

★★★ MANDATORY

Correct Answer: (B) The limit does not exist

STEP-BY-STEP SOLUTION

The quotient equals 1 for x > −2 and −1 for x < −2, so the two-sided limit does not exist.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is only the left-hand limit.

✗ **(C)** This is only the right-hand limit.

✗ **(D)** The quotient is never 0.

Question 75

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1

STEP-BY-STEP SOLUTION

For x > 0, 1/x − 1 < ⌊1/x⌋ ≤ 1/x. Multiplying by x gives 1 − x < x⌊1/x⌋ ≤ 1, and both bounds → 1.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The Squeeze Theorem gives a finite value.

✗ **(C)** The factors cancel rather than grow.

✗ **(D)** The floor term is approximately 1/x, not small.

Question 76

★★★ MANDATORY

Correct Answer: (D) $\lim_{x\to 2^-} f(x) = 1$ and $\lim_{x\to 2^+} f(x) = 2$, so $\lim_{x\to 2} f(x)$ does not exist

STEP-BY-STEP SOLUTION

Just left of 2 the floor equals 1; just right of 2 it equals 2. The unequal one-sided limits mean the limit does not exist.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The left-hand limit is 1.

✗ **(B)** This ignores the left-hand limit of 1.

✗ **(C)** The function jumps at every integer.

Question 77

★★★★★ VERY IMPORTANT

Correct Answer: (A) There is c in [0, 2] with f(c) = f(c + 2)

STEP-BY-STEP SOLUTION

Let $g(x) = f(x + 2) - f(x)$ on $[0, 2]$. Then $g(2) = f(4) - f(2) = f(0) - f(2) = -g(0)$, so $g(0)$ and $g(2)$ are zero or opposite in sign. The IVT gives a root.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The result holds for every continuous f.

✗ **(C)** The given information is sufficient.

✗ **(D)** The argument shows such a c must exist.

Question 78

★★ IMPORTANT

Correct Answer: (D) $x/|x|$

STEP-BY-STEP SOLUTION

$x/|x|$ is undefined at 0 and has one-sided limits -1 and 1.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** The exponential is continuous everywhere.

✗ **(B)** The radicand is always positive.

✗ **(C)** Absolute value is continuous everywhere.

Question 79

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1/2

STEP-BY-STEP SOLUTION

Let $u = x^2$. Then $x^4 = u^2$ and the expression becomes $(1 - \cos u)/u^2 \to 1/2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Omits the factor 1/2.

✗ **(B)** Substitutes into the numerator only.

✗ **(D)** Squares the limit value.

Question 80

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1

STEP-BY-STEP SOLUTION

Since $x \sin(1/x) \to 1$, the numerator behaves like x, so the quotient behaves like $x/(x + 1) \to 1$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses an incorrect growth comparison.

✗ **(B)** The numerator does not vanish.

✗ **(D)** One power of x cancels with $\sin(1/x)$.

Question 81

★★★ MANDATORY

Correct Answer: (C) 4/3

STEP-BY-STEP SOLUTION

Factor both: $(x - 1)(x + 1)(x^2 + 1) / [(x - 1)(x^2 + x + 1)] \to (2)(2)/3 = 4/3$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Inverts the ratio.

✗ **(B)** Confuses the x-value with the limit.

✗ **(D)** Uses only one factor of the numerator.

Question 82

★★★ MANDATORY

Correct Answer: (B) −3/2

STEP-BY-STEP SOLUTION

Factor: $(x + 1)(x^2 - x + 1) / [(x + 1)(x - 1)] = (x^2 - x + 1)/(x - 1) \to 3/(-2) = -3/2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Sign error in the denominator.

✗ **(C)** Factoring resolves the form.

✗ **(D)** Substitutes into the numerator only.

Question 83

★★★★★ VERY IMPORTANT

Correct Answer: (B) 3/2

STEP-BY-STEP SOLUTION

Let $t = (1 + x)^{(1/6)} \to 1$. The expression becomes $(t^3 - 1)/(t^2 - 1) = (t^2 + t + 1)/(t + 1) \to 3/2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Substitutes into the numerator only.

✗ **(C)** Inverts the ratio of root indices.

✗ **(D)** Assumes the roots cancel.

Question 84

★★ IMPORTANT

Correct Answer: (D) y = 2

STEP-BY-STEP SOLUTION

Equal degrees, so the asymptote is the ratio of leading coefficients, $y = 2$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses the constant terms.

✗ **(B)** The denominator does not dominate.

✗ **(C)** Uses the denominator constant.

Question 85

★★★★★ VERY IMPORTANT

Correct Answer: (A) 1

STEP-BY-STEP SOLUTION

As $x \to +\infty$ the denominator $\approx 2x$, so $f \to 1/2$. As $x \to -\infty$ the denominator $\to 0^+$ while the numerator $\to -\infty$, so f diverges. Exactly one horizontal asymptote exists.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** This overcounts the asymptotes.

✗ **(C)** The limit at $+\infty$ exists.

✗ **(D)** There is no finite limit at $-\infty$.

Question 86

★★★ MANDATORY

Correct Answer: (B) 1

STEP-BY-STEP SOLUTION

$\sqrt{(x^4 + 1)} = x^2\sqrt{(1 + 1/x^4)}$, so the quotient $= (1 + 2/x)/\sqrt{(1 + 1/x^4)} \to 1$.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses the coefficient 2 incorrectly.

✗ **(C)** Both grow at the same rate.

✗ **(D)** The denominator does not dominate.

Question 87

★★★★★ VERY IMPORTANT

Correct Answer: (A) 0

STEP-BY-STEP SOLUTION

Since $|\sqrt{x}\sin(1/x)| \le \sqrt{x} \to 0$, the Squeeze Theorem gives 0.

WHY THE OTHER OPTIONS FAIL

✗ **(B)** The damping factor forces convergence.

✗ **(C)** $\sqrt{x} \to 0$ prevents growth.

✗ **(D)** Assumes $\sin(1/x) \to 1$.

Question 88

★★★ MANDATORY

Correct Answer: (D) x = 1 and x = 2

STEP-BY-STEP SOLUTION

The floor function jumps at each integer. The integers strictly inside (0, 3) are 1 and 2.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Jumps occur at the integers.

✗ **(B)** 3 is not in the open interval.

✗ **(C)** 0 is not in the open interval.

Question 89

★★★★★ VERY IMPORTANT

Correct Answer: (D) 7

STEP-BY-STEP SOLUTION

Factor: $(x + 5)(x - 2)/(x - 2) = x + 5$, whose limit at 2 is 7. So m = 7.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** Uses the constant term 10.

✗ **(B)** Uses the constant 5 without evaluating.

✗ **(C)** Sign error on the simplified expression.

Question 90

★★★★★ VERY IMPORTANT

Correct Answer: (B) The limit does not exist

STEP-BY-STEP SOLUTION

For $x > 0$ the quotient is $x + 1 \to 1$; for $x < 0$ it is $x - 1 \to -1$. The one-sided limits differ.

WHY THE OTHER OPTIONS FAIL

✗ **(A)** This is only the right-hand limit.

✗ **(C)** Substitutes into the numerator only.

✗ **(D)** This is only the left-hand limit.

Question 91: Analyzing Limits and Continuity from a Graph

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) For $0 \le x < 2$ the graph follows $y = x + 1$, so $\lim_{x \to 2^-} f(x) = 3$.
For $2 < x \le 4$ the graph is the horizontal line $y = 3$, so $\lim_{x \to 2^+} f(x) = 3$.
Since both one-sided limits equal 3, $\lim_{x \to 2} f(x) = 3$.

(b) The graph shows a filled circle at $(2, 1)$, so $f(2) = 1$.
The limit exists and equals 3, but $3 \neq 1 = f(2)$. The third condition of continuity fails.
Therefore f is **not continuous at x = 2**, and the discontinuity is **removable**.

(c) $\lim_{x \to 4^-} f(x) = 3$ (the horizontal branch) and $\lim_{x \to 4^+} f(x) = 0$ (the branch $y = x - 4$).
Both one-sided limits are finite but unequal, so f has a **jump discontinuity** at $x = 4$.

(d) f is continuous on $[0, 2) \cup (2, 4] \cup (4, 6]$.
Note that $x = 4$ is included in the second interval because $f(4) = 3$ equals the left-hand limit.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Correct left-hand limit of 3	1 pt
(a) Correct right-hand limit of 3 and two-sided limit	2 pts
(b) States $f(2) = 1$ and compares with the limit	1 pt
(b) Correct classification as removable	1 pt
(c) Correct one-sided limits 3 and 0	1 pt
(c) Correct classification as jump	1 pt
(d) Correct intervals of continuity	2 pts

Question 92: Piecewise Function with Two Parameters

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $\lim_{x \to 1^-} f(x) = 1^2 + a = \mathbf{1 + a}$.
$\lim_{x \to 1^+} f(x) = b(1) + 2 = \mathbf{b + 2}$.

(b) The three conditions require f(1) to be defined, the limit to exist, and the two to be equal.
Here f(1) = 3 is defined. For the limit to exist we need 1 + a = b + 2, and for continuity both must equal 3.
1 + a = 3 gives **a = 2**; b + 2 = 3 gives **b = 1**.
With a = 2 and b = 1, all three conditions are satisfied.

(c) With a = 2: $\lim_{x \to 1^-} f(x) = 3$. With b = 5: $\lim_{x \to 1^+} f(x) = 7$.
The one-sided limits are both finite but $3 \neq 7$, so $\lim_{x \to 1} f(x)$ does not exist.
Therefore f has a **jump discontinuity** at x = 1.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Correct expression 1 + a	1 pt
(a) Correct expression b + 2	1 pt
(b) Sets both one-sided limits equal to f(1) = 3	2 pts
(b) Correct values a = 2 and b = 1	2 pts
(c) Evaluates one-sided limits 3 and 7	2 pts
(c) Correct classification as jump	1 pt

Question 93: Estimating a Limit from a Table of Values

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) As x approaches 3 from the left, the values 6.700, 6.970, 6.997 approach **7**.
As x approaches 3 from the right, the values 7.300, 7.030, 7.003 approach **7**.

(b) Both one-sided limits appear to equal 7, and a two-sided limit exists exactly when the one-sided limits agree.
Therefore $\lim_{x \to 3} g(x) \approx 7$.

(c) Condition 1: g(3) = 4 is defined. Condition 2: $\lim_{x \to 3} g(x) = 7$ exists.
Condition 3 fails because $7 \neq 4$.
Therefore g is **not continuous at x = 3**.

(d) Redefining **g(3) = 7** makes the function value match the limit.
This removes a **removable discontinuity** (a hole in the graph).

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Correct left-hand estimate of 7	1 pt
(a) Correct right-hand estimate of 7	1 pt
(b) States the limit is 7 with reasoning about agreeing one-sided limits	2 pts
(c) Compares g(3) = 4 with the limit of 7	2 pts
(c) Concludes not continuous	1 pt
(d) States g(3) = 7 and identifies removable discontinuity	2 pts

Question 94: Holes, Vertical Asymptotes, and End Behavior

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $x^2 - x - 6 = (x - 3)(x + 2)$ and $x^2 - 9 = (x - 3)(x + 3)$.
Cancelling the common factor gives **$h(x) = (x + 2)/(x + 3)$** for $x \neq 3$ and $x \neq -3$.

(b) ***lim*** $(x \to 3)\ h(x) = (3 + 2)/(3 + 3) =$ **5/6**.
Since the limit exists but $h(3)$ is undefined, the graph has a **hole (removable discontinuity) at (3, 5/6)**.

(c) Near $x = -3$ the numerator approaches -1 (nonzero) while the denominator approaches 0.
As $x \to -3^-$, $x + 3$ is a small negative number, so $h(x) = -1/(\text{small negative}) \to +\infty$.
As $x \to -3^+$, $x + 3$ is a small positive number, so $h(x) \to -\infty$.
The vertical asymptote is **$x = -3$**.

(d) Dividing numerator and denominator by x: $(1 + 2/x)/(1 + 3/x) \to 1$.
So ***lim*** $(x \to \infty)\ h(x) =$ **1** and the horizontal asymptote is **$y = 1$**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Correct factorization of both polynomials	1 pt
(a) Simplified form with restrictions	1 pt
(b) Correct limit 5/6	1 pt
(b) Identifies a hole at (3, 5/6)	1 pt
(c) Correct one-sided limits $+\infty$ and $-\infty$ with sign analysis	2 pts
(c) States $x = -3$	1 pt
(d) Correct limit 1 and asymptote $y = 1$	2 pts

Question 95: The Squeeze Theorem

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $\lim_{x\to 1}(4x - x^2) = 4 - 1 = \mathbf{3}$.
$\lim_{x\to 1}(x^2 + 2) = 1 + 2 = \mathbf{3}$.

(b) The inequality $4x - x^2 \le g(x) \le x^2 + 2$ holds on an open interval containing $x = 1$, and both bounding functions have the same limit 3 at $x = 1$.
By the Squeeze Theorem, $\mathbf{\lim_{x\to 1} g(x) = 3}$.

(c) At $x = 0$ the lower bound approaches $4(0) - 0 = 0$ while the upper bound approaches $0 + 2 = 2$. Because the two bounding limits are different, the Squeeze Theorem does not apply, and g could approach any value in [0, 2].

(d) For all $x \neq 0$, $-1 \le \cos(1/x) \le 1$. Multiplying by $x^2 \ge 0$ gives $-x^2 \le x^2\cos(1/x) \le x^2$.
Since $\lim_{x\to 0}(-x^2) = 0$ and $\lim_{x\to 0} x^2 = 0$, the Squeeze Theorem gives $\mathbf{\lim_{x\to 0} x^2\cos(1/x) = 0}$.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Both limits correctly evaluated as 3	2 pts
(b) States the inequality holds near $x = 1$ and both limits agree	2 pts
(b) Concludes the limit equals 3	1 pt
(c) Shows the bounding limits are 0 and 2 at $x = 0$	2 pts
(d) Correct bounds and conclusion of 0	2 pts

Question 96: The Intermediate Value Theorem with Tabular Data

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) It is given that f is continuous on the closed interval [0, 8], which is the only hypothesis required by the Intermediate Value Theorem.

(b) Examine the sign of f at consecutive table values:
$f(0) = 5 > 0$ and $f(2) = -2 < 0$, so there is a zero in $(0, 2)$.
$f(2) = -2 < 0$ and $f(4) = 3 > 0$, so there is a zero in $(2, 4)$.
$f(4) = 3 > 0$ and $f(6) = -1 < 0$, so there is a zero in $(4, 6)$.
$f(6) = -1 < 0$ and $f(8) = 4 > 0$, so there is a zero in $(6, 8)$.
Each sign change guarantees a zero by the IVT, so the minimum number is **4**.

(c) Yes. Since f is continuous on $[0, 2]$ and 1 lies strictly between $f(2) = -2$ and $f(0) = 5$, the IVT guarantees a value c in $(0, 2)$ with **$f(c) = 1$**.

(d) Yes. The IVT provides only a guaranteed minimum. The function could oscillate between the listed x-values and cross the x-axis additional times.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Cites continuity on the closed interval	2 pts
(b) Identifies all four sign changes	2 pts
(b) Correctly concludes a minimum of 4 zeros with IVT justification	2 pts
(c) States $-2 < 1 < 5$ and applies the IVT	2 pts
(d) Explains the IVT gives a minimum, not an exact count	1 pt

Question 97: Limits at Infinity and Horizontal Asymptotes

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) From the graph, g(x) levels off at **y = 2**, so ***lim*** (x→∞) g(x) = 2.
Algebraically, for x > 0 we have $\sqrt{(x^2 + 1)} = x\sqrt{(1 + 1/x^2)}$.
Therefore $g(x) = 2x / [x\sqrt{(1 + 1/x^2)}] = 2/\sqrt{(1 + 1/x^2)} \to 2/\sqrt{1} = \mathbf{2}$.

(b) From the graph, g(x) levels off at **y = −2**, so ***lim*** (x→−∞) g(x) = −2.
The key fact is that $\sqrt{(x^2)} = |x|$, and for x < 0 we have $|x| = -x$.
So $\sqrt{(x^2 + 1)} = -x\sqrt{(1 + 1/x^2)}$, giving $g(x) = 2x / [-x\sqrt{(1 + 1/x^2)}] = -2/\sqrt{(1 + 1/x^2)} \to \mathbf{-2}$.
The sign differs because the square root always returns a nonnegative value while x itself is negative.

(c) A vertical asymptote would require the denominator to equal zero. However, $x^2 + 1 \geq 1 > 0$ for every real x, so $\sqrt{(x^2 + 1)}$ is never zero.
Therefore g has **no vertical asymptotes**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Reads the limit 2 from the graph	1 pt
(a) Algebraic verification using $\sqrt{(x^2+1)} = x\sqrt{(1+1/x^2)}$	2 pts
(b) Reads the limit −2 from the graph	1 pt
(b) Uses $\|x\| = -x$ for x < 0	2 pts
(b) Explains the sign difference	1 pt
(c) Shows $x^2 + 1 > 0$ and concludes no vertical asymptotes	2 pts

Question 98: Infinite Limits and Asymptotic Behavior from a Graph ★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) From the graph, the branch to the left of x = 1 falls without bound, so $\lim_{x \to 1^-} h(x) = -\infty$.
The branch to the right of x = 1 rises without bound, so $\lim_{x \to 1^+} h(x) = +\infty$.

(b) Near x = 1 the numerator x + 1 approaches 2, which is positive and nonzero.
As $x \to 1^-$, the denominator x − 1 is a small negative number, so the quotient is (positive)/(small negative) $\to -\infty$.
As $x \to 1^+$, the denominator x − 1 is a small positive number, so the quotient is (positive)/(small positive) $\to +\infty$.

(c) The vertical asymptote is **x = 1**. Because the one-sided limits are infinite, h has an **infinite (essential) discontinuity** at x = 1.

(d) Dividing numerator and denominator by x gives $(1 + 1/x)/(1 - 1/x)$.
As $x \to \pm\infty$ both 1/x terms vanish, so both limits equal **1** and the horizontal asymptote is **y = 1**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Correct one-sided limits $-\infty$ and $+\infty$	2 pts
(b) Analyzes the sign of the numerator near x = 1	1 pt
(b) Correct sign analysis of the denominator on both sides	2 pts
(c) States x = 1 and classifies as an infinite discontinuity	2 pts
(d) Both limits equal 1 and asymptote y = 1	2 pts

Question 99: Special Trigonometric Limits and Continuity

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) Rewrite the expression so the standard limit appears:
$\sin(5x)/(3x) = (5/3) \cdot [\sin(5x)/(5x)]$.
As $x \to 0^-$ we have $5x \to 0$, so $\sin(5x)/(5x) \to 1$.
Therefore $\lim_{x \to 0^-} f(x) = (5/3)(1) = \mathbf{5/3}$.

(b) Multiply numerator and denominator by $(1 + \cos 2x)$:
$(1 - \cos 2x)/x^2 = (1 - \cos^2 2x) / [x^2(1 + \cos 2x)] = \sin^2(2x) / [x^2(1 + \cos 2x)]$.
Now write $\sin^2(2x)/x^2 = 4 \cdot [\sin(2x)/(2x)]^2 \to 4(1)^2 = 4$.
Also $1 + \cos 2x \to 1 + 1 = 2$.
Therefore $\lim_{x \to 0^+} f(x) = 4/2 = \mathbf{2}$.

(c) Continuity at $x = 0$ requires $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = f(0)$.
However $5/3 \neq 2$, so the two-sided limit does not exist for any choice of c.
Therefore **no value of c** makes f continuous at $x = 0$; the function has a jump discontinuity there.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Rewrites using the standard limit form	2 pts
(a) Correct value 5/3	1 pt
(b) Uses the conjugate or an equivalent valid method	2 pts
(b) Correct value 2	2 pts
(c) Compares 5/3 and 2 and concludes no value of c works	2 pts

Question 100: Applied Context: Cooling Model

★★★★★ VERY IMPORTANT

COMPLETE MODEL SOLUTION

(a) $T(0) = 18 + 72e^0 = 18 + 72 = \mathbf{90°C}$.
This matches the table entry of 90.0°C at t = 0.

(b) Average rate of change = $[T(20) - T(0)] / (20 - 0) = (50.4 - 90.0)/20 = -39.6/20$.
This equals **−1.98 degrees Celsius per minute**.
The negative sign indicates the coffee is cooling.

(c) As $t \to \infty$, the exponent $-0.04t \to -\infty$, so $e^{(-0.04t)} \to 0$.
Therefore $\lim_{t\to\infty} T(t) = 18 + 72(0) = \mathbf{18°C}$.
Interpretation: over a long period of time the coffee cools toward the ambient room temperature of 18°C, which it approaches but never actually reaches.

(d) T is a sum of a constant and a constant multiple of an exponential function, and both are continuous for all real t. Therefore T is continuous on [0, 60].
From the table, T(60) = 24.5 and T(0) = 90.0, and 24.5 < 40 < 90.0.
Since T is continuous on [0, 60] and 40 lies strictly between T(60) and T(0), the Intermediate Value Theorem guarantees a value t in (0, 60) with **T(t) = 40°C**.

AP SCORING CRITERIA

Evaluation Criterion	Points
(a) Computes T(0) = 90°C	2 pts
(b) Correct difference quotient setup and value −1.98 with units	2 pts
(c) Correct limit of 18°C	1 pt
(c) Correct contextual interpretation	2 pts
(d) Justifies continuity and applies the IVT with 24.5 < 40 < 90	2 pts

Essential Unit 1 Formulas and Theorems

- **Existence of a limit:** $\lim_{x\to c} f(x) = L$ if and only if $\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = L$.
- **Three conditions for continuity at c:** $f(c)$ is defined, $\lim_{x\to c} f(x)$ exists, and the two are equal.
- **Special trigonometric limits:** $\lim_{x\to 0} \sin(kx)/x = k$ | $\lim_{x\to 0} (1 - \cos x)/x = 0$ | $\lim_{x\to 0} (1 - \cos x)/x^2 = 1/2$.
- **Rational limits at infinity:** compare degrees — lower numerator gives 0, equal degrees give the leading-coefficient ratio, higher numerator diverges.
- **Radicals at negative infinity:** $\sqrt{(x^2)} = |x|$, and for $x < 0$ this equals $-x$.
- **Squeeze Theorem:** if $g(x) \le f(x) \le h(x)$ near c and $\lim g = \lim h = L$, then $\lim f = L$.
- **Types of discontinuity:** removable (limit exists but differs from the value), jump (unequal finite one-sided limits), infinite (one-sided limits unbounded).
- **Vertical asymptote:** the denominator is zero and the numerator is nonzero after all common factors cancel.
- **Intermediate Value Theorem:** if f is continuous on [a, b] and k lies between f(a) and f(b), then $f(c) = k$ for at least one c in (a, b).

Disclaimer: This resource contains original AP Calculus AB–aligned practice questions. It is not affiliated with, endorsed by, or approved by the College Board. "AP" and "Advanced Placement" are registered trademarks of the College Board.